PRE-ANALYSIS PLAN

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I. Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

I.1 Variable selection

We construct the **`has_daughter`** variable to indicate whether an individual has any daughter(s) or not during the survey year by creating a dummy indicator (0 = no daughters, 1 = with daughters). To compute this indicator, we use variables `kidgeb01` to `kidgeb19` and `kidsex01` to `kidsex19`.

We construct the dependent variable **`gender_att`** by taking the average value of the following attitude variables (all in Likert scale, 1 =strongly disagree to 4 = strongly agree, listed below):

- `lb1110`: Ein Kind unter 6 Jahren wird darunter leiden, wenn seine Mutter arbeitet
- `lb1113`: Frauen sollten sich stärker um die Familie kümmern als um ihre Karriere.
- `lb1114`: Ein Kind unter 3 Jahren wird darunter leiden wenn seine Mutter arbeitet.
- `lb1115` (reversed): Männer sollten sich genauso an der Hausarbeit beteiligen wie Frauen.
- `lb1119`: Männer sollten sich stärker um die finanzielle Absicherung der Familie kümmern als Frauen
- `lb1120` (reversed): Am besten ist es, wenn der Mann und die Frau beide gleich viel erwerbstätig sind und sich beide in gleichem Maße um Haushalt und Familie kümmern
- `lb1121` (reversed): Eine berufstätige Mutter kann Ihrem Kind genauso viel emotionale Wärme geben wie eine Mutter, die nicht arbeitet.

Regarding control variables, we pick the following variables: migration background (`migback`, recoded to dummy variable 0/1), `sampreg` (East/West Germany), **`hhinc_quant`** (the quantile of `hhinc` per survey year), years of schooling (`pgbilzeit`), marital status (`pgfamstd`), employment status (`pgemplst`), `sex` (male or not, recoded as dummy variable), age of respondent (computed from `syear` and `gebjahr`), and age of oldest child (based on variables `kidgeb01` to `kidgeb19` and `syear`).

I.2. Exclusion criteria

Only observations with answers to gender attitude-related questions are included. All observations that do not have valid values in the gender attitude variables are dropped from the analysis.

1) Exclude individuals without children (for all hypotheses): Using only parents of sons as the control group allows us to hold parenthood status constant and avoids confounding due to selection into having children.
   We also exclude one observation that reports having a total of 1 child but 3 daughters.
2) Exclude individuals whose chosen control variables are missing. In our case, this amounts only to around 5% of the entire sample, so there is no clear need to apply imputation methods.
3) Exclude all individuals who are NOT between the ages of 18 and 65.

I.3 Model specification

We run OLS regression with `gender_att` as our Y variable and X as `has_daughter`, with the control variables as outlined above. We also use robust standard errors; see the Stata code line we use below.

```
reg gender_att has_daughter male migback age_resp
i.hhinc_quant pgbilzeit i.pgfamstd i.pgemplst i.sampreg
num_children age_oldest syear, vce(robust)
```

I.4 Result interpretation

To be in line with the hypothesis, the coefficient for `has_daughter` has to be positive and the associated p-value has to be below 0.05.

II. Hypothesis 2: Having daughters leads to more left-leaning political preferences

II.1 Variable selection

`has_daughter` and control variables: same as H1.

The dependent variable used to test this hypothesis is the political preference `left_leaning` (0-10, higher values equal more left-leaning) computed by reversing the scale of the variable `plh0004`.

II.2 Exclusion criteria

Only observations with answers to political preferences are included. All observations that do not have valid values in the `left_leaning` variable are dropped from the analysis.

The exclusion criteria are the same as for Hypothesis 1, except for the years where observations are available.

II.3 Model specification

We run fixed effects regression `xtreg` with:
Y = `left_leaning(plh0004)`
X = **`has_daughter`**

The control variables are as outlined above. We also use robust standard errors. We control for the survey year and person fixed effects, see the detailed Stata code below:

```
xtset pid syear
xtreg left_leaning has_daughter age_resp i.hhinc_quant
pgbilzeit i.pgfamstd i.pgemplst i.sampreg num_children
syear, vce(robust) fe
```

I.4 Result interpretation

To be in line with the hypothesis, the coefficient for `has_daughter` has to be positive and the associated p-value has to be below 0.05.

III. Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

III.1. Variable selection

To construct the dependent variable to measure gender equality in the household, we construct a `his_housework` variable as follows:

- sum up all hours dedicated to childcare and chores spent per observation during weekdays (`childcare: pli0044_h`, `chores: pli0043_h`).
- for each respondent whose different-sex partner also answered questions `pli0044_h` and `pli0043_h` in the same survey year, compute the total hours spent on childcare and chores by the couple (hours of respondent + hours of the partner).
- divide the total hours spent on childcare and chores by *the male partner* by total hours dedicated to childcare and chores by *the couple* calculated above to get the `his_housework` variable. Thus, the final variable measures the share of a couple's housework done by the male partner.

III.2. Exclusion criteria

To measure gender equality in household labor, we restrict the sample to heterosexual couples with children, where both partners participated in the same survey year and reported their weekday hours spent on childcare and chores. Weekdays are used because they reflect routine, time-constrained days when traditional gender roles in task division are more salient, whereas weekends involve more discretionary time that may obscure persistent inequalities. Focusing on weekdays also enhances comparability across households with similar daily structures.

We exclude observations that meet any of the following criteria:

(1) the partner did not participate in the same survey year

(2) either partner did not report weekday hours on childcare and chores

(3) the couple has no children, or reports inconsistent data (e.g., one child but three daughters)

(4) control variables are missing (affecting ~5% of the sample, thus not warranting imputation)

(5) respondents are not between the ages of 18 and 65.

III.3 Model specification

We run fixed effects regression `xtreg` with:

Y = **`his_housework`**

X = **`has_daughter`**

The control variables are as outlined above. We also use robust standard errors. We control for the survey year and person fixed effects, see below the detailed Stata code line:

```
xtset pid syear
xtreg his_housework has_daughter age_resp i.hhinc_quant
pgbilzeit i.pgfamstd i.pgemplst i.sampreg num_children
syear, vce(robust) fe
```

III.4 Result interpretation

To be in line with the hypothesis, the coefficient for `has_daughter` has to be positive and the associated p-value has to be below 0.05.

IV. Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

IV.1 Variable selection

We construct the dependent variable `gender_att` by taking the average value of the attitude variables in the same fashion as Hypothesis I.

IV.2. Exclusion criteria

Same as Hypothesis I

IV.3 Model specification

We run OLS regression `reg` with robust standard errors with the same set of control variables as Hypothesis I, but including an interaction term between `has_daughter` and `male`, where:

Y = **`gender_att`**

X = **`has_daughter and c.has_daughter#c.male`**

```
reg gender_att has_daughter male c.has_daughter#c.male
migback age_resp i.hhinc_quant pgbilzeit i.pgfamstd
```

```
i.pgemplst i.sampreg num_children age_oldest syear,
vce(robust)
```

IV.4 Result interpretation

To be in line with the hypothesis, the coefficient for `c.has_daughter#c.male` has to be positive and the associated p-value has to be below 0.05.